

Reconstructed-energy background spectra for a Quantum-Parity-Detector germanium wafer: reactor CEvNS, cosmic muons, and environmental gammas

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Coherent elastic neutrino-nucleus scattering (CEvNS) from a nuclear reactor produces germanium nuclear recoils of only tens to hundreds of eV, at the threshold of emerging quantum sensors. We present an end-to-end forward model of the reconstructed-energy differential rate spectra measured by a thin (4 in. $\times$ 4 in. $\times$ 2 mm, $\sim$ 110 g) single-sided germanium wafer read out by Quantum Parity Detectors (QPDs), for two quasiparticle-trapping designs (Ta $\rightarrow$ Al and Al $\rightarrow$ Hf). On a single unified phonon energy scale with no ionization quenching, we compute the deposited-energy spectra of the reactor-CEvNS signal and of its surface backgrounds—sea-level cosmic muons, the environmental-gamma Compton continuum, the ambient fast-neutron elastic-recoil field, and the Ge-intrinsic cosmogenic decays of ^3H , ^{68}Ge , and ^{65}Zn —and fold them through a Monte-Carlo QPD response matrix $R(E_{\text{rec}} | E_{\text{dep}})$ that includes the bandwidth-limited (25 kHz) censoring of the tunneling readout. The reconstructed energy is calibrated to estimate the deposited energy, so it doubles as the physical energy an experiment would report. Every rate is anchored to a published value or evaluated data library: the CEvNS rate reproduces the Billard *et al.* (2017) germanium benchmark to within 2.4%, the integral muon rate matches the PDG sea-level flux within $\sim$ 30%, the Compton edges fall at the Klein-Nishina energies, the neutron flux is normalized to the Gordon sea-level measurement, and the cosmogenic activities to SuperCDMS and EDELWEISS germanium measurements. The resulting spectra show that the reactor-CEvNS signal reconstructs to tens of eV—but it does not occupy that band alone: the ambient neutron-elastic recoils share its kinematics and, even after an assumed shield of NUCLEUS-demonstrated performance, are the dominant background in the 10–100 eV region of interest, giving an in-band signal-to-background ratio of order a few ($S/B \simeq 2.6$); the Ge-intrinsic cosmogenic decays, which no external shield reaches, set the irreducible floor beneath it. Cosmic-muon deposits saturate the readout and pile up into a tens-of-keV feature—an instrumental artifact of the bandwidth ceiling, not a physical spectral line—well clear of the signal band. We treat the saturated-regime response, the order-of-magnitude ambient-neutron rate and its assumed suppression, the site-dependent environmental-gamma normalization, and the sub-1.8 MeV reactor flux as explicit, quantified modeling uncertainties. This forward model establishes the reconstructed-energy regime—and the neutron-limited background budget—in which a QPD reactor-CEvNS search must operate.

I. INTRODUCTION

Coherent elastic neutrino-nucleus scattering (CEvNS) offers a compact, Standard-Model-anchored probe of reactor antineutrinos, but the nuclear recoils it produces in germanium at reactor energies are only tens to hundreds of eV, below the reach of conventional ionization or phonon detectors. First predicted by Freedman [1] and proposed as a neutrino-detection channel by Drukier and Stodolsky [2], CEvNS was observed only in 2017 by the COHERENT experiment at a stopped-pion source [3], which fixed the cross-section normalization against the electroweak Standard Model. Reactor sources move the target recoil energies an order of magnitude lower, into the tens-of-eV regime, in exchange for an enormous $\bar{\nu}_e$ flux. A dedicated effort has followed—CONUS+ [4], Dresden-II [5], and NUCLEUS [6]—each pushing the deposited-energy threshold toward and below 100 eV, where the germanium reactor-CEvNS rate is

benchmarked by the calculation of Billard *et al.* [7]. The central experimental obstacle is not statistics but threshold: recovering the reactor-CEvNS spectrum demands sensitivity to nuclear recoils that terminate at a few tens of eV.

Quantum Parity Detectors (QPDs) are a superconducting readout concept aimed squarely at this regime [8]. A QPD infers a deposited energy from the population of quasiparticles it liberates in a superconducting sensor, sensed through the parity-dependent tunneling of individual quasiparticles across a Josephson junction. Because each tunneling event is counted, the scheme offers a path to sub-keV and even eV-scale thresholds on a fieldless phonon calorimeter. It carries, however, an intrinsic limitation: the tunneling signal is bandwidth-limited, and the readout response *saturates* once the instantaneous tunneling rate exceeds the resolving ceiling—here $\sim$ 25 kHz, set by a $\tau_d \simeq 40 \mu\text{s}$ dead time. A large deposit drives the burst rate far above this ceiling, so the number of *recorded* tunneling events, and hence the inferred energy, no longer tracks the energy actually deposited. The decisive observable for such a detector is therefore not the deposited energy but the *reconstructed*

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energy E_{rec} that survives the bandwidth-limited counting chain. A background that is innocuous in deposited energy can, after reconstruction, land squarely on top of an eV-scale signal.

This paper computes that reconstructed-energy observable end to end. We build a forward model that maps the deposited-energy spectrum of the reactor-CEvNS signal and of its surface backgrounds into reconstructed-energy differential rate spectra dR/dE_{rec} for a thin single-sided germanium wafer read out by QPDs. The backgrounds are of three kinds: external electron-recoil continua from through-going cosmic-ray muons and the radiogenic environmental-gamma field; the external nuclear-recoil field of ambient fast neutrons, which scatter elastically off germanium and mimic the CEvNS signature directly; and the internal Ge-intrinsic decays of the cosmogenically activated isotopes ^3H , ^{68}Ge , and ^{65}Zn , which deposit from inside the crystal and which no external shield can reach. We treat two quasiparticle-trapping designs, Ta $\rightarrow$ Al and Al $\rightarrow$ Hf [8], and we place every channel on a single unified phonon energy scale with no ionization quenching, as befits a fieldless calorimeter, so that signal and backgrounds are directly co-added and folded through a common response. Every rate is anchored to a published value or evaluated data library rather than assumed: the deposited CEvNS rate reproduces the Billard germanium benchmark to within 2.4%, the cosmic-muon flux matches the sea-level value tabulated by the Particle Data Group to within its $\sim 30\%$ inter-experiment spread [9], the environmental-gamma Compton edges are fixed exactly by scattering kinematics, the ambient neutron flux is anchored to the Gordon sea-level measurement [10], and the cosmogenic production rates to the SuperCDMS and EDELWEISS germanium-activation measurements [11, 12]. Because the external channels can be attenuated but the internal floor cannot, we present the background budget in an assumed-shield scenario—external muon, gamma, and neutron fields suppressed at the performance level demonstrated by NUCLEUS [6], the Ge-intrinsic floor carried unsuppressed. The response chain that carries these deposited spectra into E_{rec} —the sensor-sharing partition, the tunneling-burst statistics, and the non-paralyzable censoring at the 25 kHz ceiling—is the methodological core of the work.

The headline finding is a qualitative separation on the reconstructed axis. Because the reconstructed energy is calibrated to *estimate* the deposited energy, the reactor-CEvNS signal, whose recoils lie entirely below the saturation onset, reconstructs to tens of eV, with a median near 55 eV and about two-thirds of its counts below 100 eV. It reconstructs onto the very lowest end of the observable axis—but it does not sit there alone. The ambient neutron-elastic recoils occupy the same tens-of-eV region, and even after the assumed shield they are the largest term in the reconstructed 10–100 eV region of interest, at $\sim 4.8 \times 10^3$ counts $\text{kg}^{-1}\text{day}^{-1}$ against a CEvNS signal of ~ 63 : the flagship channel is signal-limited by

a nuclear-recoil background that shares its kinematics, not by the readout. The cosmic-muon channel, by contrast, deposits ~ 1.5 –197 MeV and drives the readout deep into saturation, so its entire deposited spectrum collapses onto a single tens-of-keV plateau (≈ 37.6 keV for Ta $\rightarrow$ Al, ≈ 29.9 keV for Al $\rightarrow$ Hf), well clear of the signal band. Beneath everything lies the Ge-intrinsic cosmogenic floor, whose ^{68}Ge and ^{65}Zn electron-capture lines reconstruct into the shoulder just above the RoI and whose ^3H continuum underlies it. We stress at the outset what this separation is and is not. It is a stage-1 forward-model, or feasibility, calculation, not a sensitivity or discovery claim, and we quote no detection significance. The muon pile-up is an *instrumental artifact* of the modeled bandwidth ceiling, not a physical spectral line, and the shape of the saturated-regime response has no published anchor; the neutron channel carries only an order-of-magnitude accuracy label and its assumed suppression is a modeling input, not a measured rejection; the absolute environmental-gamma normalization is a site-dependent estimate good to a factor of ~ 2 ; and the crossover between the linear and saturated regimes rides on two low-confidence sensor-sharing parameters, so we report it as a band spanning roughly 50 eV to 13 keV rather than a single number. These modeling uncertainties are quantified channel by channel throughout, so that the separation stands on stated assumptions rather than hidden ones.

The remainder of the paper follows the pipeline in order. Section II fixes the detector geometry, the unified phonon energy scale, and the per-sensor tunneling chain. Section III builds the deposited reactor-CEvNS signal from an absolute antineutrino flux and the coherent cross section, and anchors it to the Billard benchmark. Section IV computes the deposited cosmic-muon, environmental-gamma, ambient-neutron, and Ge-intrinsic cosmogenic spectra. Section V constructs the bandwidth-limited response matrix $R(E_{\text{rec}} | E_{\text{dep}})$ that is the heart of the forward model. Section VI folds every channel through it to obtain the reconstructed spectra of Fig. 1 in the assumed-shield scenario, and Secs. VII and VIII discuss the implications and limits of the result.

II. DETECTOR MODEL AND ENERGY SCALE

The detector is a single-sided germanium wafer, 4 in. $\times$ 4 in. $\times$ 2 mm, of density $\rho = 5.323$ g cm^{-3} , giving a volume of ~ 20.6 cm^3 and a mass of ~ 110 g. At the natural-abundance molar mass this corresponds to 8.29×10^{24} germanium atoms per kilogram; we quote all differential rates per unit target mass (counts $\text{kg}^{-1}\text{day}^{-1}$ keV^{-1}) so they can be compared directly with the reactor-CEvNS benchmark literature, while the physical target for absolute rates is the 110 g wafer. One face carries $\sim 10,300$ Quantum Parity Detector (QPD) sensors at a 1 mm pitch, each a superconducting quasiparticle

trap read out through a charge-parity junction [8]. We consider the two trapping designs of that work, Ta→Al and Al→Hf, whose device parameters are taken verbatim from its Table II and summarized in Table I. The two designs differ only in the trap material and its junction parameters; they share the geometry, the energy scale, and the reconstruction chain developed below.

Unified phonon energy scale. A fieldless phonon calorimeter has no ionization-collection channel and no Luke–Neganov gain, so the full recoil energy of a deposit thermalizes into the phonon system regardless of whether the primary recoil is nuclear (CEvNS) or electronic (cosmic-muon ionization, Compton). We therefore place every channel on a single unified phonon energy scale with *no* ionization quenching: a nuclear recoil and an electron recoil of equal deposited energy E_{dep} land at the same point on the observable axis. Applying a Lindhard or ionization-yield factor here would suppress the CEvNS rate by $\sim 5\text{--}7\times$ and would be physically incorrect for this sensor [1]; we consequently never mix keV_{ee} and keV_{nr} scales. The phonon energy is $E_{\text{ph}} = E_{\text{dep}} - E_{\text{stored}}$, where the defect (Frenkel-pair) storage E_{stored} is a few-percent nuclear-recoil-only correction and is zero for muon and Compton deposits; we take $E_{\text{ph}} \simeq E_{\text{dep}}$ throughout.

Deposit-to-signal chain. The phonon energy is partitioned among the sensors (the localized-plus-diffuse partition, which sets the per-sensor share E_s , is specified in Sec. V). A per-sensor share E_s breaks Cooper pairs to yield a trapped-quasiparticle count N_{qp} , a density n_{qp} over the effective trap volume V_{tr} , and a plateau tunneling rate Γ_{in} set by the per-design tunneling constant K :

$$\begin{aligned} N_{\text{qp}} &= \varepsilon \frac{E_s}{\Delta_{\text{tr}}}, & n_{\text{qp}} &= \frac{N_{\text{qp}}}{V_{\text{tr}}}, \\ \Gamma_{\text{in}} &= K n_{\text{qp}}, & \Gamma_{\text{in}}^{\text{pk}} &= p \Gamma_{\text{in}}, \end{aligned} \quad (1)$$

where Δ_{tr} is the trap superconducting gap and ε the deposited-to-signal efficiency (below). The steady rate $\Gamma_{\text{in}} = K n_{\text{qp}}$ is dimensionally consistent (K in $\text{Hz } \mu\text{m}^3$, n_{qp} in μm^{-3}), and $N_{\text{qp}} = \varepsilon E_s / \Delta_{\text{tr}}$ is a pure count. The instantaneous burst peaks above its plateau by a two-exponential pulse peak factor p (from the injection/recombination profile of Ref. [8], Eq. 4), giving the peak rate $\Gamma_{\text{in}}^{\text{pk}}$ that governs saturation.

Efficiency. We impose $\varepsilon \approx 0.5$ as a forward-model definition lumping phonon collection, pair-breaking, and trapping; it sets the quasiparticle yield $N_{\text{qp}} = \varepsilon E_s / \Delta_{\text{tr}}$ and therefore enters the saturation onset and the trigger. It is a physical deposit-to-signal conversion fraction, *not* the energy scale: the reconstructed energy is calibrated to estimate the deposited energy (Sec. V), which absorbs ε into the count-to-energy constant so that $E_{\text{rec}} \approx E_{\text{dep}}$ in the unsaturated regime, as any on-line calibration against a known deposit would. This is a definitional baseline, not a derived value; we carry a design-dependent $\pm 10\text{--}20\%$ band on it and adopt the same central value for both designs. It is distinct from, and should not be conflated

TABLE I. Trap device parameters for the two designs, from Table II of Ref. [8]. Δ_{tr} is the trap gap, V_{tr} the effective trap volume, K the tunneling constant ($\Gamma_{\text{in}} = K n_{\text{qp}}$), and p the two-exponential peak factor.

Parameter	Ta→Al	Al→Hf
Trap material	Al	Hf
Δ_{tr} (μeV)	190	40
V_{tr} (μm^3)	100	1000
K ($\text{kHz } \mu\text{m}^3$)	3	20
p	0.25	0.13

with, the paper’s physical efficiency estimate $\eta_{\text{ce}} \approx 0.3$, which we retain only as an independent cross-reference.

Bandwidth-limited readout and saturation. The tunneling readout is sampled at 50 kHz (interval 20 μs), which by the Nyquist criterion resolves tunneling transients up to 25 kHz, i.e. a resolving (dead) time $\tau_d = 40 \mu\text{s}$. A deposit saturates the readout when its peak tunneling rate exceeds this ceiling,

$$\Gamma_{\text{in}}^{\text{pk}} > \Gamma_{\text{max}} \equiv \frac{1}{\tau_d} = 25 \text{ kHz}, \quad \tau_d = 40 \mu\text{s}. \quad (2)$$

Because Δ_{tr} , V_{tr} , K , and p differ between designs, the two designs reach this ceiling at different deposit energies; the Al→Hf design saturates first (its plateau Γ_{in} per unit deposit is $\sim 3.2\times$ that of Ta→Al). The per-design crossover deposit energy and the whole-array plateau are quoted in Sec. V. The mapping from the true tunneling rate Γ to the observed count rate m is a dead-time censoring model, and the choice of model is not fixed by electronics reasoning alone; both the non-paralyzable and paralyzable idealizations are defensible:

$$\begin{aligned} m &= \frac{\Gamma}{1 + \Gamma \tau_d} \quad (\text{non-paralyzable}), \\ m &= \Gamma e^{-\Gamma \tau_d} \quad (\text{paralyzable}). \end{aligned} \quad (3)$$

The non-paralyzable form is monotone and saturates at $m \rightarrow 1/\tau_d = 25 \text{ kHz}$; the paralyzable form is non-monotone, peaking at $\Gamma = 1/\tau_d$ and rolling over thereafter. This was an open modeling choice, closed by decision: we adopt *non-paralyzable* censoring as the canonical convention for all results and retain the paralyzable variant only as a sensitivity. Equations (1)–(3) fix the setup on which the flux and background channels (Secs. III and IV) and the reconstruction estimator (Sec. V) are built.

III. REACTOR ANTINEUTRINO FLUX AND CEVNS RATE

The reactor antineutrino signal enters our pipeline as a deposited nuclear-recoil spectrum dR/dT , where $T \equiv E_{\text{nr}}$ is the germanium recoil kinetic energy on the unified

phonon scale of Sec. II. We build it in three steps: an absolute antineutrino flux $\Phi(E_\nu)$, the coherent-scattering cross section $d\sigma/dT$, and their fold over the five natural germanium isotopes. All rates in this section are on the *deposited* energy T ; the mapping to reconstructed energy E_{rec} is deferred to Secs. V and VI.

A. Reactor antineutrino flux

We take the emitted per-fission spectra $S_i(E_\nu)$ from the Huber conversion fits [13] for ^{235}U , ^{239}Pu , and ^{241}Pu and from the Mueller fit [14] for ^{238}U above 2 MeV, with coefficients taken from the primary arXiv e-prints; our reconstructed ^{235}U spectrum agrees with the published Huber tabulation to 0.73% at 3 MeV and 0.13% at 5 MeV. Below 1.8 MeV, where the conversion fits do not extend, we seam-match (continuous through the derivative) a summation-style continuation plus a $^{238}\text{U}(n, \gamma)$ component normalized to 0.6 $\bar{\nu}_e$ per fission. The absolute flux at the detector is

$$\Phi(E_\nu) = \frac{R_f}{4\pi d^2} \left[\sum_i f_i S_i(E_\nu) + S_{n\gamma}(E_\nu) \right], \quad (4)$$

$$R_f = \frac{P_{\text{th}}}{\langle E_f \rangle},$$

where R_f is the fission rate, $P_{\text{th}} = 3 \text{ GW}_{\text{th}}$ is the *thermal* reactor power, $\langle E_f \rangle = 205.8 \text{ MeV}$ is the fission-fraction-weighted effective thermal energy per fission (not the total Q -value, which double-counts the escaping neutrinos), f_i are the fission fractions, and $d = 25 \text{ m}$ is the standoff, treated as a point source. This gives $R_f = 9.10 \times 10^{19} \text{ fissions s}^{-1}$ and an integrated flux $\int \Phi dE_\nu = 7.5 \times 10^{12} \bar{\nu}_e \text{ cm}^{-2}\text{s}^{-1}$. The corresponding emission of $1.96 \times 10^{20} \bar{\nu}_e \text{ s}^{-1} \text{ GW}_{\text{th}}^{-1}$ reproduces the Hayes–Vogel reactor-accounting value $\sim 2 \times 10^{20}$ [15]. The sub-1.8 MeV shape is a model placeholder consistent with, but not directly digitized from, Kopeikin [16]; we carry it with a wide below-2-MeV uncertainty band and quantify its downstream impact at the end of this section.

B. Coherent cross section

For each isotope i we use the Freedman coherent cross section [1, 2],

$$\frac{d\sigma_i}{dT} = \frac{G_F^2 M_i}{4\pi} Q_{W,i}^2 \left(1 - \frac{M_i T}{2E_\nu^2} \right) F_i^2(q), \quad (5)$$

with the $/4\pi$ prefactor (the $/8\pi$ form is a factor-of-two error), G_F the Fermi constant, M_i the nuclear mass, and weak charge $Q_{W,i} = N_i - (1 - 4\sin^2 \theta_W)Z$ evaluated at the low-energy value $\sin^2 \theta_W = 0.2387$. Because $1 - 4\sin^2 \theta_W = 0.045$, the proton coupling nearly vanishes and the rate scales as N^2 to $\sim 0.5\%$. The momentum

transfer is $q = \sqrt{2M_i T}$ and $F_i(q)$ is the Lewin–Smith Helm form factor [17, 18], which stays above 0.998 at reactor recoil energies on germanium. The cross section is evaluated in natural units and restored to cm^2 with $(\hbar c)^2 = 3.894 \times 10^{-28} \text{ GeV}^2 \text{ cm}^2$; dropping this factor is the single most common normalization error in reactor CEvNS. Equation (5) integrates to the closed form $\sigma_{\text{tot}} = G_F^2 Q_W^2 E_\nu^2 / 4\pi$ to better than 6×10^{-8} , and yields $\sigma(^{72}\text{Ge}, 4 \text{ MeV}) = 1.0026 \times 10^{-40} \text{ cm}^2$, within 0.26% of the first-principles anchor. The Helm parameters and the per-isotope germanium table are collected in Sec. A.

C. Deposited-energy rate

The differential rate follows from folding Eq. (4) against Eq. (5) over the five stable germanium isotopes, retaining each isotope’s own weak charge, nuclear mass, and kinematic threshold rather than a lumped- A approximation:

$$\frac{dR}{dT} = N_T \sum_i x_i \int_{E_\nu^{\text{min}}(T)}^\infty \Phi(E_\nu) \frac{d\sigma_i}{dT} dE_\nu, \quad (6)$$

$$E_\nu^{\text{min}}(T) = \sqrt{\frac{1}{2} M_i T},$$

where $N_T = 8.29 \times 10^{24} \text{ Ge atoms kg}^{-1}$ is the target density, x_i the natural isotopic abundances (neutron numbers $N_i = 38, 40, 41, 42, 44$), and $E_\nu^{\text{min}}(T) = \sqrt{M_i T/2}$ is the minimum antineutrino energy that produces a recoil T . Integrating Eq. (6) gives the deposited spectrum dR/dT shown in Fig. 4, with 67.8 counts $\text{kg}^{-1}\text{day}^{-1}$ above a 50 eV_{nr} deposited threshold at 3 GW_{th} and 25 m. Propagating the split flux band through Eq. (6) yields a 1σ uncertainty of 3.4–10% on the deposited rate, dominated at the lowest recoils by the sub-1.8 MeV placeholder: it contributes 18–34% of the rate below 95 eV_{nr}, corresponding to a ~ 6 –10% band there, while the well-anchored $> 1.8 \text{ MeV}$ flux controls the rate at every higher recoil.

D. Absolute-normalization benchmark

The decisive external check is a reproduction of the Billard *et al.* germanium reactor-CEvNS rates [7]. Their Table 1 is quoted for a 8.54 GW_{th} source at $\sim 400 \text{ m}$, so we rescale a flux variant built under their assumptions (Huber–Mueller held constant below 2 MeV, no n -capture) to that configuration by the geometry-and-power factor

$$k = \frac{P_B}{P_{\text{th}}} \left(\frac{d}{d_B} \right)^2 = 0.01112, \quad (7)$$

with $P_B = 8.54 \text{ GW}_{\text{th}}$ and $d_B \simeq 400 \text{ m}$; both powers are thermal, so only the $1/d^2$ geometry survives in

the ratio. Single-source and explicit two-core evaluations of Eq. (7) agree to 0.25%. Folding with k reproduces the Billard rates 0.76/0.51/0.26 counts $\text{kg}^{-1}\text{day}^{-1}$ above 50/100/200 eV_{nr} as 0.742/0.501/0.257, within 2.4%. Omitting k overshoots by exactly $1/k = 89.9\times$ in every bin, confirming that the residual mismatch is purely the geometry-and-power normalization and not the spectral shape or the cross section. As a second, coarser anchor, rescaling both our flagship prediction and the Billard rate to the CONUS+ configuration [4] agrees to a factor 0.99, though this comparison is at the ionization (eV_{ee}) scale and requires a quenching model to relate to our deposited phonon scale, so we treat it as a rate-scale cross-check only. Together these anchors fix the absolute normalization of the deposited CEvNS signal that Secs. V and VI carry through the QPD response.

IV. COSMIC-MUON AND ENVIRONMENTAL-GAMMA BACKGROUNDS

A surface germanium wafer sits in several ambient radiation fields. Two are electron-recoil continua from external radiation—through-going cosmic-ray muons and the radiogenic environmental-gamma continuum—and a third is the nuclear-recoil field of ambient fast neutrons, which elastically scatter off germanium and are the background most dangerous to a CEvNS search because they mimic its very signature. A fourth field is internal: cosmogenic activation during surface exposure seeds the crystal with the radioisotopes ^3H , ^{68}Ge , and ^{65}Zn , whose decays deposit energy from *inside* the target. We compute each deposited spectrum dR/dE_{dep} on the unified phonon scale (no ionization quenching), and we anchor each to a published benchmark or evaluated data library so that the downstream reconstructed spectra rest on validated inputs rather than assumed shapes.

The four fields differ in how they respond to shielding, and this organizes the scenario we present. The external muon, gamma, and neutron fields can be attenuated by a passive shield and active veto; we adopt the performance demonstrated by NUCLEUS [6] as a concrete assumed shield, with suppression factors of 1000 for muons, 200 for neutrons, and 500 for environmental gammas, and a boron-loaded moderator that also removes the ambient thermal-neutron flux and hence the in-situ (n, γ) capture and ^{71}Ge channels it would otherwise drive. The internal cosmogenic decays, by contrast, occur within the crystal and *no external shield reaches them*: they are the irreducible floor. The reconstructed background budget of Sec. VI is therefore the shielded ambient channels stacked on the unsuppressed Ge-intrinsic floor. We present the per-channel forward model at its unshielded rate below and apply the assumed suppression only at the point of assembly, so that the suppression factors remain explicit, separable modeling inputs rather than buried assumptions.

A. Cosmic muons

We sample the sea-level muon intensity from the modified-Gaisser (“Gaisser–Guan”) parametrization [19, 20], which incorporates the low-energy/large-angle correction that plain Gaisser lacks,

$$\frac{dI}{dE_\mu d\Omega} = 0.14 \left(\frac{E_\mu}{\text{GeV}} \right)^{-2.7} \times \left[\frac{1}{1 + \frac{1.1 E_\mu \cos \theta^*}{115 \text{ GeV}}} + \frac{0.054}{1 + \frac{1.1 E_\mu \cos \theta^*}{850 \text{ GeV}}} \right], \quad (8)$$

in $\text{cm}^{-2}\text{s}^{-1}\text{sr}^{-1}\text{GeV}^{-1}$, where E_μ is the muon energy and the effective zenith angle $\cos \theta^* = [(\cos^2 \theta + P_1^2 + P_2 \cos^3 \theta + P_4 \cos^5 \theta)/(1 + P_1^2 + P_2 + P_4)]^{1/2}$ softens the horizontal divergence of the plain $\sec \theta$ scaling (P_1 – $P_5 = 0.102573, -0.068287, 0.958633, 0.0407253, 0.817285$).

Each muon piercing the wafer traverses a chord ℓ fixed by the ray–box intersection of its straight trajectory with the $10.16 \times 10.16 \times 0.20$ cm slab. The chord ranges from the 0.20 cm vertical thickness to the 14.37 cm space diagonal, with the isotropic-flux mean pinned by Cauchy’s theorem to $\langle \ell \rangle = 4V/S = 0.385$ cm. The deposited spectrum is the flux-weighted, projected-area chord distribution folded with the straggling density,

$$\frac{dR}{dE_{\text{dep}}} = \frac{1}{m} \int d\Omega (\hat{n} \cdot \hat{\Omega})_+ A \int dE_\mu \frac{dI}{dE_\mu d\Omega} \times f_L(E_{\text{dep}}; \Delta_p(\ell, E_\mu), \xi(\ell)), \quad (9)$$

where m is the wafer mass, A the entry-face area, $(\hat{n} \cdot \hat{\Omega})_+$ the inward surface projection, and f_L the Landau–Vavilov density whose mode sits at the *most-probable value* (MPV) Δ_p . Using the mean deposit here, or the Moyal approximation, would misplace the peak and suppress the high-energy tail; we use neither. With mass thickness $x = \rho\ell$, the width and MPV follow the standard forms

$$\xi = \frac{K}{2} \frac{Z}{A} \frac{x}{\beta^2}, \quad \Delta_p = \xi \left[\ln \frac{2m_e c^2 \beta^2 \gamma^2}{I} + \ln \frac{\xi}{I} + j - \beta^2 - \delta(\beta\gamma) \right], \quad (10)$$

with $K = 0.307 \text{ MeV mol}^{-1}\text{cm}^2$, $Z/A = 0.4406$ for Ge, mean excitation $I \approx 350 \text{ eV}$, $j = 0.200$, and $\delta(\beta\gamma)$ the density-effect correction. For a vertically incident minimum-ionizing muon ($x = 1.065 \text{ g cm}^{-2}$) we obtain $\xi = 0.072 \text{ MeV}$, matching the PDG value [9], and $\Delta_p = 1.23 \text{ MeV}$ — strictly below the mean deposit $\langle \Delta \rangle = 1.46 \text{ MeV}$. This ordering $\Delta_p < \langle \Delta \rangle$ is the signature of the strongly right-skewed Landau distribution and is the single most consequential modeling choice in this channel.

Rare near-horizontal trajectories traverse up to the full 14.37 cm diagonal and deposit up to $\sim 197 \text{ MeV}$; we importance-sample these long chords so this high-deposit tail is statistically resolved. It is the tail — not the $\sim \text{MeV}$

bulk — that drives the bandwidth-limited readout into saturation, as analyzed in Sec. V. Integrating Eq. (9) over flux and projected area gives a total muon rate through the wafer of 1.366 Hz. This is consistent with the PDG sea-level expectation ($\sim 1 \text{ muon cm}^{-2} \text{ min}^{-1}$, $I_v \approx 70 \text{ m}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$) to within $\sim 20\%$, well inside the VALD-02 validation tolerance of 30%; the residual reflects the known 30–35% inter-experiment normalization spread of the Gaisser–Guan flux and is the weakest anchor of this otherwise textbook-grade channel.

B. Environmental gammas

The wafer sits in a radiogenic gamma field sourced by ^{40}K and the ^{238}U and ^{232}Th decay chains. The line energies are fixed nuclear data; we take the absolute per-line fluxes from the LABChico measured survey [21] (for example, the ^{40}K 1461 keV line at $0.036 \text{ cm}^{-2} \text{ s}^{-1}$), with the line list following the standard low-background compilation [22]. Because the 2 mm wafer is optically thin to MeV photons — the interaction probability is $\mu\bar{\ell} \approx 0.117$ per mean chord at 1 MeV, so the double-scatter fraction is only $\sim 1.4\%$ — each photon Compton-scatters at most once and the scattered photon escapes. The deposit is therefore the recoil-electron kinetic energy $T_e = E_\gamma - E'$, a *continuum*, and not a full-energy photopeak.

We sample the scattering angle θ from the bound-electron incoherent cross section,

$$\frac{d\sigma}{d\Omega} = \frac{r_e^2}{2} \left(\frac{E'}{E_\gamma} \right)^2 \left(\frac{E'}{E_\gamma} + \frac{E_\gamma}{E'} - \sin^2 \theta \right) S(x, Z), \quad (11)$$

$$E' = \frac{E_\gamma}{1 + \alpha(1 - \cos \theta)},$$

with classical electron radius r_e , incident photon energy E_γ , scattered energy E' , and $\alpha = E_\gamma/m_e c^2$. The Klein–Nishina factor [23] sets the continuum shape and the free-electron edge; the Hubbell incoherent scattering function [24] $S(x, Z=32)$, evaluated at the momentum-transfer variable

$$x = \frac{E_\gamma [\text{keV}] \sin(\theta/2)}{12.39842} \text{ \AA}^{-1}, \quad (12)$$

imposes electron binding: since $S(x \rightarrow 0) \rightarrow 0$, it suppresses near-forward (low-recoil) scattering, rolling the deposited continuum off near 30–50 eV and removing the unphysical free-Klein–Nishina low-energy plateau, while $S(x \rightarrow \infty) \rightarrow Z$ leaves the bulk Compton continuum unchanged. The recoil deposit for each line terminates at the Compton edge,

$$E_{\text{edge}} = \frac{2E_\gamma^2}{m_e c^2 + 2E_\gamma}, \quad (13)$$

which places the ^{40}K , ^{214}Bi , and ^{208}Tl edges at 1243, 1541, and 2382 keV respectively. These edge positions are the sharpest, most model-independent features of the

channel: they are fixed by Compton kinematics alone and are reproduced by the angle-sampled maxima to better than 0.5 keV.

Summed over lines, the single-scatter interaction rate is 0.267 Hz, within 2% of the independent $\Phi \sigma_{\text{KN}} N_e$ estimate (0.273 Hz, ratio 0.98), where Φ is the integral gamma flux, σ_{KN} the Klein–Nishina cross section, and N_e the wafer electron number. We are explicit about the split in confidence: the edge positions are exact (HIGH confidence), whereas the absolute normalization inherits the site-dependent gamma flux and is a representative estimate good to a factor of ~ 2 (MEDIUM confidence). The ^{238}U -chain flux in particular is set by an assumed chain balance rather than a directly measured line intensity, and a different deployment site would rescale the overall rate — but not the edge structure — accordingly.

C. Ambient neutron elastic recoils

Fast neutrons are the background most dangerous to a CEvNS search: they scatter elastically off germanium nuclei and produce exactly the low-energy nuclear recoils the signal consists of, on the same phonon scale and with no discriminating handle. We take the sea-level ambient neutron field from the PARMA/EXPACS analytical model [25], evaluated at a representative mid-latitude, sea-level, ground-level point, and normalize it to the $> 10 \text{ MeV}$ integral flux measured by Gordon *et al.* [10], $\Phi(> 10 \text{ MeV}) = 3.55 \times 10^{-3} \text{ cm}^{-2} \text{ s}^{-1}$; the un-normalized PARMA prediction already agrees with that anchor to $\sim 9\%$. Only this single integral is externally validated, and we attach an order-of-magnitude accuracy label to the entire channel accordingly—the eV–keV differential shape that actually sets the in-band recoil rate is not independently benchmarked, and the local low-energy flux is moreover sensitive to the moderating and reflecting material around the detector at the tens-of-percent level.

The deposited recoil spectrum is the elastic fold

$$\frac{dR}{dT} = N_{\text{Ge}} \int_{E_{\min}(T)}^{E_{\text{top}}} \phi(E_n) \frac{\sigma_{\text{el}}(E_n)}{f E_n} dE_n, \quad E_{\min} = \frac{T}{f}, \quad (14)$$

with σ_{el} the natural-germanium elastic cross section from ENDF/B-VIII.0 [26], N_{Ge} the target density, and $f = \langle 4A/(1+A)^2 \rangle = 0.0536$ the abundance-weighted kinematic factor relating a recoil T to the incident neutron energy. Because the epithermal flux is nearly $\phi \propto 1/E_n$, the integrand behaves as $1/E_n^2$ and is peaked at its lower limit $E_{\min}(T)$, so the recoil rate in the 10–100 eV region of interest is set by the neutron flux at $E_{\min} = T/f \approx 0.19$ – 1.9 keV —the epithermal band, precisely the part of the spectrum most sensitive to the local environment. The resolved ^{73}Ge resonance at $E_n = 102.59 \text{ eV}$ imprints a kinematic edge at $T = f E_n \simeq 5.5 \text{ eV}_{\text{dep}}$ (reconstructing to $\sim 5.3 \text{ eV}$): a neutron on that resonance can transfer at most 5.5 eV, so the resonance’s large contribution switches off sharply above it. Folded through the re-

sponse, the *unshielded* channel integrates to 4.78×10^3 (Ta $\rightarrow$ Al) and 4.83×10^3 counts $\text{kg}^{-1}\text{day}^{-1}$ in the reconstructed 10–100 eV RoI. This is the largest background in the band by nearly two orders of magnitude, and it remains the dominant term even after the assumed neutron suppression factor of 200.

D. Ge-intrinsic cosmogenic activation

Cosmic-ray spallation during above-ground exposure activates the germanium itself, seeding it with radioisotopes that then decay from *inside* the crystal. Three dominate the sub-keV budget: ^3H , ^{68}Ge (decaying through ^{68}Ga), and ^{65}Zn . We take their sea-level production rates from the SuperCDMS/CDMSlite [11] and EDELWEISS-III [12] measurements and evaluate the as-deployed activity for a fixed one-year surface-exposure scenario with no cool-down (a continuously operating surface run, where activation keeps building), giving 4.05, 18.22, and 10.98 decays $\text{kg}^{-1}\text{day}^{-1}$ for ^3H , ^{68}Ge , and ^{65}Zn respectively; the longer exposures allowed by the production-rate band raise these by up to $\sim 3\times$. Because these deposits are born inside the target, they carry no quenching and *no external shield attenuates them*—they are the irreducible floor of the shielded scenario.

The ^3H decay is a β^- continuum with an 18.6 keV endpoint; we use the allowed spectral shape with the non-relativistic Fermi function, so its deposited spectrum is a smooth continuum from threshold to 18.6 keV, of which the fraction falling in the RoI is small but nonzero. The ^{68}Ge and ^{65}Zn decays proceed by electron capture, depositing the full atomic de-excitation energy (X-ray plus Auger cascade) of the daughter shell vacancy as discrete lines: the K, L, and M shells of the ^{68}Ga daughter at 10.37 keV, 1.30 keV, and 158.7 eV, and of the ^{65}Cu daughter at 8.98 keV, 1.10 keV, and 120 eV. On the calibrated reconstructed axis these lines image near their deposited energies, only mildly compressed by saturation above the onset: the ^{68}Ga M line at 158.7 eV reconstructs to ~ 124 eV and its L line to ~ 0.87 keV, so the sub-keV EC lines sit in the *shoulder* just above the 100 eV RoI rather than inside it. Only the low-energy tail of the ^3H continuum and the M-shell lines approach the band, but as an unshielded internal floor they set the ultimate reach of the scenario once the ambient channels are suppressed.

All background spectra are computed on the shared phonon energy grid, so they are co-added—the ambient muon, gamma, and neutron channels under the assumed shield suppression, the Ge-intrinsic $^3\text{H}/^{68}\text{Ge}/^{65}\text{Zn}$ floor unsuppressed—and folded through the detector response of Sec. V to yield the reconstructed background budget of Fig. 1. There the neutron-elastic channel dominates the RoI, the muon high-deposit tail is the agent of read-out saturation, and the cosmogenic floor sets the residual once the shield is applied.

V. QPD RESPONSE AND ENERGY RECONSTRUCTION

The device model of Sec. II fixed the per-sensor chain from a sensor share E_s to the peak tunneling rate $\Gamma_{\text{in}}^{\text{pk}}$ and the saturation ceiling $\Gamma_{\text{max}} = 1/\tau_d = 25$ kHz [Eqs. (1)–(3)], but deferred two ingredients to this section: how the deposited energy E_{dep} is partitioned among the $\sim 10,300$ sensors, and how a reconstructed energy E_{rec} is estimated from the resulting censored counts. We specify both here and use them to build the Monte-Carlo response matrix $R(E_{\text{rec}} | E_{\text{dep}})$ that Sec. VI folds through the deposit spectra. This is the methodological core of the pipeline, and also its principal source of model risk: the saturated-regime response has no published benchmark, and the crossover between the linear and saturated regimes rides on two low-confidence sharing parameters. We are explicit about both below.

Sensor-sharing partition. A single deposit does not illuminate the $N_{\text{sens}} \simeq 10,300$ sensors uniformly. We adopt a localized-plus-diffuse partition: a prompt fraction f_{prompt} of E_{dep} is absorbed near the impact point and spread over a spot of $\sim \pi r^2$ sensors, while the remaining $(1 - f_{\text{prompt}})$ thermalizes diffusely across the whole array. The on-spot and off-spot per-sensor shares are then

$$\begin{aligned} E_s^{\text{on}} &= \frac{f_{\text{prompt}} E_{\text{dep}}}{\pi r^2} + \frac{(1 - f_{\text{prompt}}) E_{\text{dep}}}{N_{\text{sens}}}, \\ E_s^{\text{off}} &= \frac{(1 - f_{\text{prompt}}) E_{\text{dep}}}{N_{\text{sens}}}, \end{aligned} \quad (15)$$

where r is the spot radius in sensor-pitch units. Both sides carry units of energy. The two parameters f_{prompt} (default 0.3, range 0.1–0.5) and r (default 2, range 1–5) have no direct thin-wafer QPD measurement; we treat them as exposed, scanned inputs rather than derived values, and we carry the equal-split limit $f_{\text{prompt}} \rightarrow 0$ (uniform sharing) as a required sanity bound. Because E_s^{on} is the largest per-sensor share, the on-spot sensors are the first to saturate, and f_{prompt} and r set the deposit energy at which that happens.

Count-integral estimator. On each sensor the trapped-quasiparticle population $N_{\text{qp}}(E_s)$ [Eq. (1)] drives a two-exponential tunneling burst. The quantity that controls both the burst statistics and the saturation is the expected number of *tunneling events*, $n_{\text{ev}} = \int \Gamma_{\text{in}}(t) dt = K \tau_{\text{qp}} N_{\text{qp}} / V_{\text{tr}}$, where τ_{qp} is the trapped-quasiparticle lifetime; this is distinct from, and much smaller than, the trapped count N_{qp} itself (by ~ 0.03 for Al, ~ 0.008 for Hf), and it is n_{ev} , not N_{qp} , that we feed as the Poisson mean to the exponentially-modified-Gaussian (EMG) tunneling-burst template of Ref. [8]. Conflating the two would mis-scale the entire saturation. The template realizes the event train, which we censor with the dead-time model $m(\Gamma_{\text{in}})$ [Eq. (3)] to obtain a per-sensor observed count $N_{\text{obs},i}$; summing over the array gives the recon-

structed energy

$$E_{\text{rec}} = C N_{\text{obs}}, \quad N_{\text{obs}} = \sum_{i=1}^{N_{\text{sens}}} N_{\text{obs},i}, \quad (16)$$

$$N_{\text{obs},i} = \int m(\Gamma_{\text{in},i}(t)) dt.$$

The count-to-energy constant C (units eV per event) is a *single global* constant per design, fixed once on a deeply unsaturated deposit by requiring the linear slope $dE_{\text{rec}}/dE_{\text{dep}} = 1$. It is not tuned per energy bin. The reconstructed energy is thus an *estimator of the deposited energy*: a detector is calibrated on a known line, and that calibration absorbs the physical deposit-to-quasiparticle conversion fraction $\varepsilon \simeq 0.5$ (Sec. II) into C , so that a 10 eV deposit reconstructs at ~ 10 eV rather than at $\varepsilon \times 10$ eV. The efficiency ε retains its physical role upstream—in the quasiparticle yield $N_{\text{qp}}(E_s)$, the saturation onset, and the trigger—but it is *not* the energy scale. Consequently, the recovery of $E_{\text{rec}} \approx E_{\text{dep}}$ below the saturation onset is a calibration-consistency check, not an independent validation; the genuinely predictive content of Eq. (16) is the saturated-regime behavior, where the estimator departs from the deposit and which no calibration can undo, to which we turn next.

Saturation and the crossover band. When the on-spot peak rate reaches the ceiling, $\Gamma_{\text{in}}^{\text{pk}} = \Gamma_{\text{max}}$ [Eq. (2)], the censored count $N_{\text{obs},i}$ stops tracking n_{ev} : under non-paralyzable censoring it *plateaus* at $\sim 1/\tau_d$ per sensor, so the summed count in Eq. (16) plateaus and E_{rec} saturates. This plateau *is* the modeled bandwidth saturation; at no point do we linearly extrapolate a rate to an energy. The retained paralyzable variant instead rolls the count over (the observed rate peaks at $\Gamma = \Gamma_{\text{max}}$ and declines), which we carry only as a sensitivity. Writing E_{onset} for the per-sensor share at which $\Gamma_{\text{in}}^{\text{pk}} = \Gamma_{\text{max}}$ (~ 1.27 eV for Ta $\rightarrow$ Al, ~ 0.77 eV for Al $\rightarrow$ Hf, from Eqs. (1)–(2)), the crossover deposit energy at which the on-spot sensor first saturates follows from inverting Eq. (15):

$$E_{\text{dep}}^{\text{cross}} = \frac{E_{\text{onset}}}{\frac{f_{\text{prompt}}}{\pi r^2} + \frac{1 - f_{\text{prompt}}}{N_{\text{sens}}}}. \quad (17)$$

Because $E_{\text{dep}}^{\text{cross}}$ inherits the f_{prompt}/r uncertainty, we report it as a band rather than a single number. At the default point it is ~ 52.9 eV (Ta $\rightarrow$ Al) and ~ 32.1 eV (Al $\rightarrow$ Hf); over the full $f_{\text{prompt}} \in [0.1, 0.5]$, $r \in [1, 5]$ scan it ranges up to ~ 13 keV and ~ 7.9 keV in the equal-split limit, with a whole-array plateau (every sensor saturated) at ~ 18.6 keV and ~ 11.3 keV. The Al $\rightarrow$ Hf design saturates before Ta $\rightarrow$ Al at every point of the scan, consistent with its lower per-sensor onset. These are the SIMU-01 design numbers forward-referenced in Sec. II; the spread of roughly three orders of magnitude is f_{prompt} -dominated and is an honest reflection of the sharing uncertainty, not a computed precision.

Response matrix. We assemble the bandwidth-limited transfer function

$$R(E_{\text{rec}} | E_{\text{dep}}) = \left. \frac{dP}{dE_{\text{rec}}} \right|_{E_{\text{dep}}}, \quad (18)$$

$$\int R(E_{\text{rec}} | E_{\text{dep}}) dE_{\text{rec}} = 1,$$

the normalized conditional distribution of reconstructed energy for a fixed deposit, per design and per censoring variant. Each E_{dep} column is built by Monte-Carlo, histogramming $N_s = 5000$ forward realizations of Eq. (16) over a grid of 584 deposit energies sampled uniformly in $\log E_{\text{dep}}$ from 10.14 eV to 197 MeV (80 bins per decade), spanning the sub-keV CEvNS floor through the deep-saturated cosmic-muon tail. The $\sim \pi r^2$ on-spot sensors are realized individually with the EMG template; the $\sim 10,287$ identical off-spot sensors are drawn as one aggregate ensemble per realization rather than looped. Columns normalize to unity to 10^{-12} , and the well-populated cells converge to $\lesssim 1.8\%$ at $N_s = 5000$ with clean $1/\sqrt{N_s}$ scaling. Figure 2 shows the resulting median $E_{\text{rec}}(E_{\text{dep}})$ with its 16–84% band for both designs and both variants. At the 197 MeV muon endpoint the non-paralyzable response plateaus at ~ 69.5 keV (Ta $\rightarrow$ Al) and ~ 53.7 keV (Al $\rightarrow$ Hf)—some three to four orders of magnitude below the linear $E_{\text{rec}} = E_{\text{dep}}$ line (197 MeV)—while the paralyzable variant rolls over to ~ 6.5 keV and ~ 5.2 keV. No column ramps linearly into the MeV range.

Limiting-case validation and its limits. The estimator is validated only at its two limits (VALD-04): $E_{\text{rec}} \approx E_{\text{dep}}$ below the crossover, and a plateau once the peak tunneling rate exceeds 25 kHz, driven by the censoring model rather than by any numerical artifact (with censoring disabled, the count integral returns to exactly E_{dep} , so the plateau demonstrably has teeth). We state plainly that *there is no literature anchor for the shape of the response between these two limits*: the saturated-regime curve is a model prediction, not a benchmarked result. The tens-of-keV muon pile-up that this section predicts is therefore an instrumental artifact of the modeled 25 kHz ceiling—not a physical spectral feature—and should be read as “the model says,” not “the detector measures.” With this caveat in hand, Eq. (18) is the object we fold the deposited-energy spectra through in Sec. VI.

VI. RECONSTRUCTED-ENERGY SPECTRA

We now fold the deposited-energy spectra of Secs. III and IV through the non-paralyzable response matrix $R(E_{\text{rec}} | E_{\text{dep}})$ of Sec. V to obtain the reconstructed-energy differential rate spectra dR/dE_{rec} for the two trapping designs, in the assumed-shield scenario of Sec. IV: the external muon, gamma, and neutron channels enter suppressed at the NUCLEUS-demonstrated performance level, and the Ge-intrinsic cosmogenic floor

enters unsuppressed. This is the payoff of the forward model: the reconstructed axis is what a bandwidth-limited readout actually records, and it differs qualitatively from the deposited axis wherever the tunneling burst saturates the 25 kHz ceiling. The fold is a discrete matrix product,

$$\left. \frac{dR}{dE_{\text{rec}}} \right|_j = \frac{1}{\Delta E_{\text{rec},j}} \sum_i R_{ji} N_{\text{dep},i}, \quad (19)$$

$$N_{\text{dep},i} = \left. \frac{dR}{dE_{\text{dep}}} \right|_i \Delta E_{\text{dep},i},$$

where $N_{\text{dep},i}$ is the rate in deposited bin i , R_{ji} is the probability that a deposit in bin i is reconstructed into bin j , and $\Delta E_{\text{dep},i}$, $\Delta E_{\text{rec},j}$ are the bin widths. Because every column of R is normalized to unit sum, Eq. (19) conserves counts per channel by construction; we verify this numerically to a relative $\sim 10^{-16}$ for every channel and both designs. The CEvNS and neutron-elastic spectra, computed on recoil grids in Secs. III and IV C, are first rebinned onto the shared E_{dep} grid by a piecewise log-log power-law integral that conserves counts to 9.6×10^{-5} overall. Because the reconstructed energy is calibrated to estimate the deposit (Sec. V), a sub-onset deposit is reconstructed via the linear mapping $E_{\text{rec}} \approx E_{\text{dep}}$ rather than discarded, and carried into the matching E_{rec} bins.

Figure 1 is the central result: dR/dE_{rec} versus reconstructed energy for the reactor-CEvNS signal and every background channel—ambient neutron elastic, cosmic muons, environmental gammas, and the Ge-intrinsic ${}^3\text{H}/{}^{68}\text{Ge}/{}^{65}\text{Zn}$ floor—with the external channels shown at their assumed-shield suppression. The reactor-CEvNS signal is drawn unsuppressed; the grey band marks the reconstructed saturation region. Four features organize the spectrum.

First, and decisively, the reactor-CEvNS signal does not have the tens-of-eV region to itself. The signal, whose recoils sit entirely below the saturation onset, reconstructs to tens of eV: its differential rate has a median near 55 eV in E_{rec} , with about two-thirds of its counts below 100 eV, and it integrates to $\sim 1.19 \times 10^2$ counts $\text{kg}^{-1}\text{day}^{-1}$, of which 62.6 (Ta $\rightarrow$ Al) and 63.6 (Al $\rightarrow$ Hf) fall in the 10–100 eV region of interest. But the ambient neutron-elastic recoils populate exactly the same band, and even after the assumed neutron suppression factor of 200 they integrate to 23.9 and 24.2 counts $\text{kg}^{-1}\text{day}^{-1}$ in the RoI—the single largest background term, $\sim 99\%$ of the residual budget. The resulting in-band signal-to-background ratio is $S/B \simeq 2.6$ for both designs, i.e. the signal exceeds the summed background by a factor of a few. This is the load-bearing result of the paper, and it is the reverse of what a deposited-threshold argument alone would suggest: recovering reactor CEvNS is limited not by the readout but by a nuclear-recoil background that shares the signal’s kinematics, and the achievable S/B is set by how well the ambient neutron field can be shielded and known. We attach an order-of-magnitude label to the neutron rate,

so $S/B \simeq 2.6$ should be read as “of order a few,” not as a percent-level number.

Second, the cosmic-muon deposits all reconstruct to a single saturation plateau, well clear of the signal band. Muon energy depositions span roughly 1.5 MeV to ~ 197 MeV (Sec. IV); every one drives the peak tunneling rate orders of magnitude above the 25 kHz ceiling, so the count-integral estimator saturates and the entire channel piles up at $E_{\text{rec}} \approx 37.6$ keV (Ta $\rightarrow$ Al) and ≈ 29.9 keV (Al $\rightarrow$ Hf). We emphasize that this tens-of-keV feature is an *instrumental artifact of the bandwidth ceiling, not a physical spectral line*: the deposited muon spectrum is broad and structureless, and the pile-up appears only because the saturating response compresses three-to-four orders of magnitude of deposited energy onto the bounded plateau. The saturated-regime response shape has no literature anchor at any energy and is validated by limiting cases only (Sec. V); the muon channel lies almost entirely within this regime, so its plateau energy must be read as a modeled ceiling. Under the assumed muon veto it contributes negligibly to the RoI.

Third, the environmental-gamma Compton continuum fills the region between the two, with reconstructed peaks near 33.5 keV (Ta $\rightarrow$ Al) and 26.6 keV (Al $\rightarrow$ Hf) and an unsuppressed integrated rate of 2.1×10^5 counts $\text{kg}^{-1}\text{day}^{-1}$; the incoherent-scattering binding suppression (Sec. IV) rolls the deposited continuum off near 30–50 eV E_{dep} , so no hard cut appears at the grid edge, and under the assumed lead shielding it too is negligible in the RoI.

Fourth, beneath the shielded ambient channels lies the Ge-intrinsic cosmogenic floor, which no shield attenuates. The ${}^{68}\text{Ga}$ and ${}^{65}\text{Cu}$ electron-capture lines image near their deposit energies: the ${}^{68}\text{Ga}$ M line at 158.7 eV reconstructs to ~ 124 eV—in the shoulder just *above* the RoI—while the ${}^{65}\text{Cu}$ M line at 120 eV reconstructs to ~ 96 eV, just inside it, and the L and K lines of both stand at ~ 0.75 – 0.87 and ~ 4.3 – 4.8 keV. Together with the low-energy tail of the ${}^3\text{H}$ β continuum, the intrinsic floor contributes ~ 0.1 – 0.2 counts $\text{kg}^{-1}\text{day}^{-1}$ to the RoI—two orders of magnitude below the shielded neutron term today, but the irreducible limit that would remain if the ambient neutron field were shielded away entirely. It sets the ultimate floor of the scenario and can be lowered only by minimizing surface exposure, not by shielding.

The saturating muon and Compton channels dominate the raw event rate, which raises the question of whether reconstruction remains operable. The total in-wafer event rate is 1.634 Hz (1.366 Hz muon + 0.267 Hz Compton, before veto), giving a mean inter-event time of 0.61 s, vastly longer than the ~ 40 μs pulse. The resulting pileup occupancy is $R\tau \approx 3.3 \times 10^{-5}$ at the 20 μs (50 kHz) sampling interval (6.5×10^{-5} at the 40 μs resolving time). Quiescent reconstruction is therefore *not* precluded, and the occupancy remains well below unity even at ten times the assumed flux.

The two designs give qualitatively similar spectra but

differ in where saturation sets in. Because the Hf trap has the larger tunneling response per unit deposit, it saturates earlier and plateaus at a lower reconstructed energy than Ta $\rightarrow$ Al, shifting the saturated features (muon pile-up, Compton peak, whole-array plateau) downward in E_{rec} ; below the crossover the two coincide on the linear $E_{\text{rec}} \approx E_{\text{dep}}$ mapping, and their in-RoI S/B agree to within the order-of-magnitude label. The shaded bands in Fig. 1 propagate the dominant systematics through Eq. (19): the CEvNS 1σ reactor-flux band (3.4% above 95 eV_{nr}, rising to 6.2%/9.8% at 50/20 eV_{nr}), the order-of-magnitude neutron band, the environmental-gamma factor-of-two site-flux band, and the muon $\pm 30\%$ normalization band. Of these it is the neutron band, not the readout, that dominates the uncertainty on S/B .

VII. DISCUSSION

The reconstructed spectra of Fig. 1 are best read not as a rate prediction but as a map of *where*, on the axis a bandwidth-limited QPD readout actually records, the reactor signal and each ambient background come to rest. That map is the deliverable of this work: it tells a would-be experiment which instrumental knob is decisive before any exposure is accumulated. Its interpretation, however, is bounded by five modeling uncertainties of very different character and provenance, and an honest reading requires keeping them separate.

Ambient neutron rate and its suppression. The uncertainty that dominates the headline result is the one on the background that dominates the RoI. The ambient fast-neutron flux carries only an order-of-magnitude accuracy label: its > 10 MeV integral is anchored to a measurement, but the eV–keV differential shape that sets the in-band recoil rate is not independently benchmarked and is moreover sensitive at the tens-of-percent level to the moderating material around the detector. On top of this, the factor-200 neutron suppression is an *assumed* shield performance taken from NUCLEUS, not a rejection we compute for this geometry. The reported $S/B \simeq 2.6$ is therefore “of order a few” and could move by an order of magnitude in either direction; it is a statement about the scenario, not a measured ratio. What would tighten it is concrete: an in-situ neutron survey at the deployment site, an as-built shield simulation, and a reactor-neutron transport calculation for the specific core and standoff. Of the five, this is the uncertainty a real experiment must retire first.

Saturated-regime response shape. The uncertainty with the least external support acts on the muon and high-recoil channels. Between the linear regime ($E_{\text{rec}} \approx E_{\text{dep}}$) and the fully saturated plateau, the shape of $R(E_{\text{rec}} | E_{\text{dep}})$ has no published anchor at any energy; it is fixed only by its two limiting cases and by calibration consistency (Sec. V). The direct consequence is that the tens-of-keV muon pile-up of Fig. 1 is an *instrumental artifact* of the modeled 25 kHz ceiling rather than

a physical spectral line. What would tighten it is unambiguous: a direct QPD saturation measurement on a thin wafer, or a full microphysical readout simulation resolving the tunneling-burst statistics through the censoring transition. Crucially, this caveat is quarantined to the channels that actually saturate—muons and the high-recoil gamma tail—because the reactor-CEvNS recoils sit entirely within the linear regime and never test the unbenchmarked shape.

Environmental-gamma normalization. The absolute Compton rate is a representative, site-dependent estimate good to a factor of ~ 2 , driven by the assumed radiogenic flux and, in particular, by the ^{238}U -chain balance rather than a measured line intensity (Sec. IV). The Compton edge positions, by contrast, are fixed by kinematics and carry high confidence. A factor-of-two shift rescales the continuum but preserves both the edge structure and the qualitative conclusion that the gamma continuum overlaps the signal region, so the *ranking* of this background is robust even though its normalization is not. Only an on-site gamma survey at the intended deployment can replace the representative estimate with a measured one.

Sub-1.8 MeV reactor flux. Below the reach of the Huber–Mueller conversion fits, the antineutrino spectrum is a cited but not directly reproduced model placeholder, carried with a wide band that translates into a ~ 6 – 10% uncertainty on the reconstructed CEvNS rate below ~ 95 eV (Sec. III). This is the leading *signal-side* systematic, and it acts precisely where the reconstructed CEvNS spectrum peaks. It is, however, a band on the rate, not an ambiguity in the spectral placement: the signal remains anchored at tens of eV regardless of where in the band the true low-energy flux lies. Digitizing a perisotope summation or CONFLUX table below 1.8 MeV, following the approach of Ref. [16], would collapse this band to the percent level.

Crossover deposit energy. The linear-to-saturated crossover rides on the two low-confidence sensor-sharing parameters f_{prompt} and r , for which no thin-wafer QPD measurement exists; we therefore report it as a band spanning ~ 50 eV to ~ 13 keV rather than a single number (Sec. V). Because the CEvNS recoils fall below even the lowest end of this band, the signal reconstructs linearly across the entire scan, so its placement is insensitive to where the true crossover sits; the uncertainty is inherited only by the saturated backgrounds, whose pile-up energies shift with the band. A dedicated thin-wafer sensor-sharing measurement would fix f_{prompt} and r and turn the band into a line.

Read together, the uncertainties separate cleanly by what they touch. The neutron rate governs the *signal-to-background* result and is the one to retire first; the saturated-response and crossover uncertainties act only on the muon and gamma backgrounds, spectrally displaced from the signal; and the gamma-normalization and low-energy-flux bands rescale rates without moving the signal off its tens-of-eV placement. The flagship qual-

itative statement—that reactor CEvNS reconstructs to tens of eV, into a band it shares with the ambient neutron recoils—is the most robust feature of the analysis. This is the sense in which the reconstructed axis is informative despite the model risk.

The implications for a QPD reactor-CEvNS search follow directly and are mostly about detector design rather than statistics. First, *threshold placement is decisive*: the CEvNS signal populates the very lowest reconstructed energies, so recovering it demands an eV-scale reconstruction floor, not merely a low deposited threshold—a requirement that places the signal below the sub-100 eV regimes now being probed by cryogenic low-threshold detectors [6] and reactor-CEvNS experiments generally. Second, cosmic muons do *not* preclude quiescent operation: their pile-up lands at tens of keV, well *above* the CEvNS band, and the pileup occupancy is tiny ($R\tau \approx 3 \times 10^{-5}$), so muon events are sparse and spectrally displaced from the signal. A surface deployment would nonetheless benefit from muon tagging or an active veto, since the rare long-chord muon tail is the very driver of readout saturation and any misreconstruction leaks downward in energy. Third, and most limiting for a signal search, the ambient neutron-elastic recoils sit *directly on top of the signal*: they share its nuclear-recoil kinematics, populate the same tens-of-eV band, and—even under the assumed factor-200 shield—are the dominant term in the RoI at $\sim 99\%$ of the residual background, giving $S/B \simeq 2.6$. Unlike the muon and gamma continua, which are spectrally displaced or shieldable to negligibility, the neutron background is irreducible by kinematics alone; suppressing it—passive moderation and capture, an active neutron veto, and overburden—is the primary lever for turning this forward model into a viable search. Beneath even a perfect neutron shield lies the Ge-intrinsic cosmogenic floor, which no external shield reaches and which minimizing surface exposure before deployment is the only way to lower. These conclusions place the QPD concept squarely within the reactor-CEvNS program pioneered at the ionization scale by CONUS+ [4], the Dresden-II point-contact germanium result [5], and the germanium rate benchmarks of Ref. [7]; the distinction here is the unified phonon scale and the bandwidth-limited reconstruction, both inherited from the QPD device concept of Ref. [8].

We close by stating the scope plainly. This is a stage-one forward-model feasibility study: it establishes the reconstructed-energy regime in which a QPD germanium wafer would operate and identifies the design choices that regime forces. It does *not* compute a sensitivity, and it makes no claim of detectability or discovery significance. Converting this map into a sensitivity would require an exposure model, a noise-and-resolution treatment of the reconstruction floor, and—above all—the measured inputs that the five caveats above call for: an in-situ neutron survey with an as-built shield and reactor-neutron transport calculation, a QPD saturation calibration, an on-site gamma survey, a digitized low-energy reactor flux,

and a thin-wafer sensor-sharing measurement. Each is a concrete, bounded next step, and together they define the path from the present feasibility map to a quantitative reactor-CEvNS reach—one that, on the evidence here, will be set by the neutron background above all.

VIII. CONCLUSIONS

We have built and executed an end-to-end, published-value-anchored forward model of the reconstructed-energy background spectra for a thin single-sided germanium QPD wafer. Starting from the deposited-energy spectra of the reactor-CEvNS signal and of its surface backgrounds—through-going cosmic muons, the radiogenic environmental-gamma continuum, the ambient fast-neutron elastic-recoil field, and the Ge-intrinsic cosmogenic decays of ^3H , ^{68}Ge , and ^{65}Zn —we placed every channel on a single unified phonon energy scale with no ionization quenching, and folded them through a common bandwidth-limited response matrix $R(E_{\text{rec}} | E_{\text{dep}})$ for two quasiparticle-trapping designs, Ta $\rightarrow$ Al and Al $\rightarrow$ Hf. The reconstructed energy is calibrated to estimate the deposit, so the output dR/dE_{rec} is both what a 25 kHz readout records and the physical energy an experiment would report.

Four findings organize the result. The reactor-CEvNS signal reconstructs to tens of eV, with a median near 55 eV in E_{rec} and about two-thirds of its counts below 100 eV—but it does not occupy that band alone. The ambient neutron-elastic recoils share its kinematics and, even under an assumed shield of NUCLEUS-demonstrated performance, are the dominant background in the 10–100 eV region of interest at $\sim 99\%$ of the residual budget, giving an in-band signal-to-background ratio of order a few, $S/B \simeq 2.6$ for both designs. This is the load-bearing conclusion: the QPD reactor-CEvNS search is neutron-limited, not readout-limited. Cosmic muons, depositing ~ 1.5 –197 MeV, drive the readout deep into saturation and collapse onto a single tens-of-keV pile-up (≈ 37.6 keV for Ta $\rightarrow$ Al, ≈ 29.9 keV for Al $\rightarrow$ Hf)—an instrumental artifact of the modeled ceiling, not a physical line—well clear of the signal band, with a pileup occupancy of only $R\tau \approx 3.3 \times 10^{-5}$ so that quiescent reconstruction is not precluded. Beneath the shielded ambient channels, the Ge-intrinsic cosmogenic decays—which no external shield reaches—set the irreducible floor. Every channel is anchored: the deposited CEvNS rate reproduces the Billard germanium benchmark to within 2.4% [7], the muon flux matches the Particle Data Group sea-level value within its $\sim 30\%$ spread, the Compton edges are fixed exactly by scattering kinematics, the neutron flux is normalized to the Gordon sea-level measurement [10], and the cosmogenic activities to the SuperCDMS and EDELWEISS germanium-activation measurements [11, 12].

We state the scope plainly. This is a stage-one forward-model feasibility study. It establishes the reconstructed-

energy regime in which a QPD germanium wafer would operate; it does *not* compute a sensitivity and makes no claim of detectability or discovery significance. Five modeling uncertainties bound the interpretation. The one that governs the headline S/B is the ambient neutron rate—order-of-magnitude by label, with an *assumed* rather than computed shield and reactor-correlated neutrons omitted—so $S/B \simeq 2.6$ should be read as “of order a few.” The remaining four act away from the signal-to-background result: the unbenchmarked saturated-regime response shape and the crossover deposit energy (a ~ 50 eV–13 keV band) touch only the muon and gamma backgrounds, while the factor-of- ~ 2 site-dependent gamma normalization and the cited-but-not-reproduced sub-1.8 MeV reactor flux rescale rates without moving the signal off its tens-of-eV placement.

Each caveat maps to a concrete next step. An in-situ neutron survey, an as-built shield simulation, and a reactor-neutron transport calculation would replace the assumed suppression and are the first to retire; a direct QPD saturation calibration would replace the modeled ceiling shape; an on-site gamma survey would fix the absolute Compton normalization; a thin-wafer sensor-sharing measurement would turn the crossover band into a line; and a digitized low-energy reactor flux would collapse the signal-side band. Combined with an exposure model and a noise-and-resolution treatment of the reconstruction floor, these define the path from the present feasibility map to a quantitative QPD reactor-CEvNS sensitivity—one that, on the evidence here, will be set by the neutron background above all—extending the phonon-scale reach of the QPD concept [8] into the reactor-antineutrino program.

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Appendix A: Cross-section and form-factor conventions

This appendix collects the cross-section and form-factor detail deferred from Sec. III: the full Freedman differential cross section with its normalization conventions, the closed-form total against which the differential is checked, the Lewin–Smith Helm form factor and its parameters, and the per-isotope natural-germanium table that enters the fold of Eq. (6).

1. Freedman cross section and closed-form check

For a neutrino of energy E_ν scattering coherently off isotope i with recoil energy $T \equiv E_{\text{nr}}$, we use the Freed-

man standard-model cross section [1, 2]

$$\frac{d\sigma_i}{dT} = \frac{G_F^2 M_i}{4\pi} Q_{W,i}^2 \left(1 - \frac{M_i T}{2E_\nu^2}\right) F_i^2(q) (\hbar c)^2, \quad (\text{A1})$$

with the Fermi constant G_F , nuclear mass M_i , and weak charge $Q_{W,i} = N_i - (1 - 4\sin^2\theta_W)Z$ evaluated at $\sin^2\theta_W = 0.2387$ (low-energy $\overline{\text{MS}}$ value, not the M_Z value 0.2312) and $Z = 32$. Two normalization choices are load-bearing and are the most common reactor-CEvNS errors: the $/4\pi$ prefactor (the $/8\pi$ form is a factor-of-two error), and the explicit conversion $(\hbar c)^2 = 3.894 \times 10^{-28} \text{ GeV}^2\text{cm}^2$ that restores the natural-unit cross section (evaluated with G_F in GeV^{-2} , M_i and E_ν in GeV) to cm^2 . Because $1 - 4\sin^2\theta_W = 0.0452$ the proton coupling nearly vanishes and $Q_{W,i}^2$ scales as N_i^2 to $\sim 0.5\%$. The kinematic factor $(1 - M_i T/2E_\nu^2)$ enforces the endpoint $T_{\text{max}}^{(i)}(E_\nu) = 2E_\nu^2/(M_i + 2E_\nu)$, and each fold in Eq. (6) runs above $E_\nu^{\text{min},(i)}(T) = \frac{1}{2}(T + \sqrt{T^2 + 2M_i T}) \simeq \sqrt{M_i T/2}$.

Setting $F_i = 1$ and integrating Eq. (A1) over $0 \leq T \leq T_{\text{max}}$ uses $\int_0^{T_{\text{max}}} (1 - M_i T/2E_\nu^2) dT = E_\nu^2/M_i$, giving the closed-form total

$$\sigma_{\text{tot},i}(E_\nu) = \frac{G_F^2 Q_{W,i}^2 E_\nu^2}{4\pi} (\hbar c)^2. \quad (\text{A2})$$

Our differential reproduces Eq. (A2) to a relative 3×10^{-9} – 6×10^{-8} across the five isotopes and $E_\nu \in \{2, 4, 8\}$ MeV, and yields $\sigma(^{72}\text{Ge}, 4 \text{ MeV}) = 1.0026 \times 10^{-40} \text{ cm}^2$, 0.26% from the first-principles $1.0 \times 10^{-40} \text{ cm}^2$ anchor. Dimensionally $[G_F^2 M_i E_\nu^2] = \text{GeV}^{-1}$, and $(\hbar c)^2$ carries the remaining GeV^2cm^2 , so $d\sigma/dT$ is in cm^2/GeV with $Q_{W,i}^2$, the kinematic factor, and F^2 dimensionless.

2. Lewin–Smith Helm form factor

The nuclear form factor is the Lewin–Smith parametrization [17] of the Helm ansatz [18],

$$F(q) = \frac{3j_1(qR_0)}{qR_0} \exp\left(-\frac{1}{2}q^2s^2\right), \quad (\text{A3})$$

$$R_0^2 = c^2 + \frac{7}{3}\pi^2a^2 - 5s^2,$$

with j_1 the first spherical Bessel function, diffraction radius $c = 1.23 A^{1/3} - 0.60$ fm, surface diffuseness $a = 0.52$ fm, and skin thickness $s = 0.90$ fm; $F(0) = 1$ by construction. The momentum transfer $q = \sqrt{2M_i T}$ is expressed in fm^{-1} via $q[\text{fm}^{-1}] = q[\text{MeV}]/(\hbar c)$ with $\hbar c = 197.327 \text{ MeV fm}$, so that qR_0 and qs are dimensionless (feeding q in the wrong units produces a large spurious suppression). For germanium at reactor recoils the form factor is nearly unity: $F^2 > 0.998$ in the sub-100 eV region that carries most of the rate, falling to $F^2(^{72}\text{Ge}) = 0.9963$ at $T = 200$ eV and to 0.9639 only at the rare ~ 2 keV endpoint reached by $E_\nu \sim 8$ –10 MeV neutrinos. These are consistent: the > 0.998

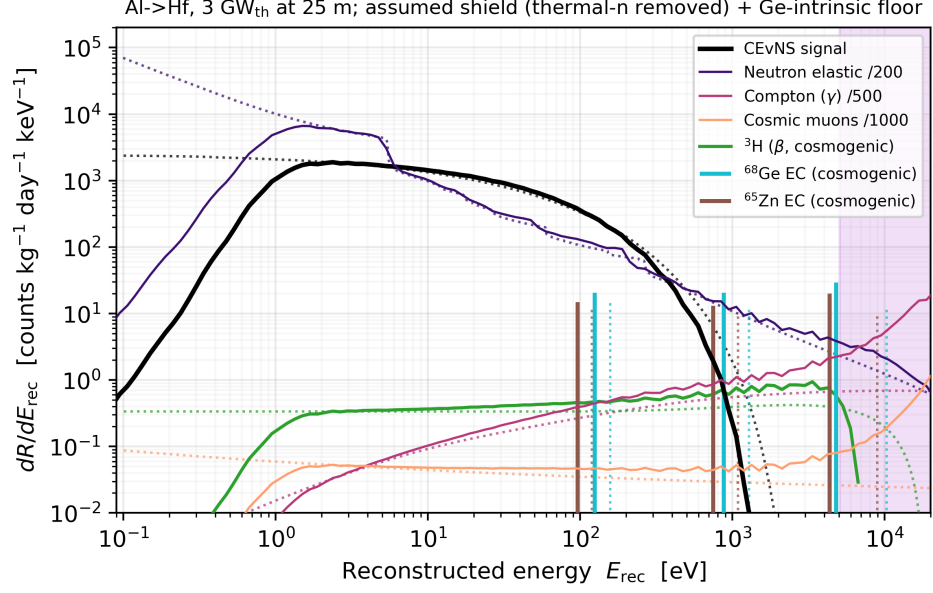


FIG. 1. Central result: reconstructed-energy differential rate spectra dR/dE_{rec} (counts $\text{kg}^{-1} \text{day}^{-1} \text{keV}^{-1}$) for the Al→Hf design under non-paralyzable censoring, in the assumed-shield scenario (Ta→Al is qualitatively identical). The reconstructed energy is calibrated to estimate the deposited energy. The reactor-CEvNS signal (black, thick) reconstructs to tens of eV. The ambient neutron-elastic recoils (purple) occupy the same band and, even after the assumed neutron suppression factor of 200 shown, are the dominant background there; cosmic-muon deposits (orange, suppressed by the assumed factor 1000 veto) span ~ 1.5 –197 MeV yet pile up at tens of keV—an instrumental artifact of the bandwidth ceiling, not a physical line; the environmental-gamma Compton continuum (magenta, suppressed by the assumed factor 500) fills the region between. The Ge-intrinsic cosmogenic floor— ^3H (green), ^{68}Ge and ^{65}Zn electron-capture lines (cyan and brown stems)—is drawn unsuppressed, as no external shield reaches it. Solid curves are the reconstructed (saturated, triggered) spectra; dotted curves are the corresponding linear-response, pre-saturation forms. Shaded purple marks the bandwidth-saturated region. The suppression factors are the assumed-shield modeling inputs of Sec. IV, quoted in the legend.

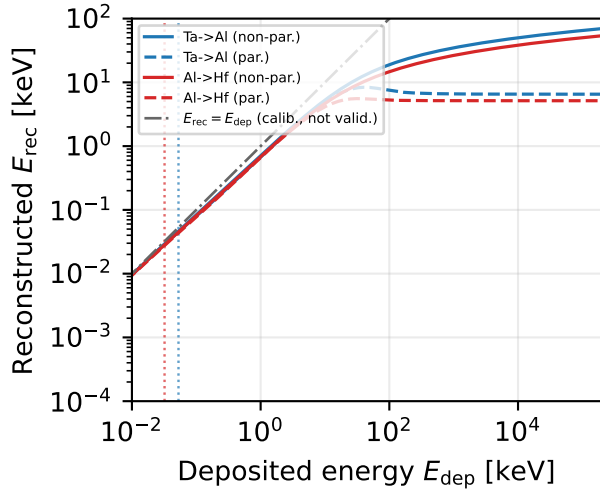


FIG. 2. Monte-Carlo QPD response $R(E_{\text{rec}} | E_{\text{dep}})$ for both designs and both censoring variants, showing the linear $E_{\text{rec}} \approx E_{\text{dep}}$ regime (the reconstructed energy calibrated to estimate the deposit), the crossover onset, and the saturation plateau.

figure is the dominant low-recoil regime and the ~ 0.96 value is the high-recoil tail. The integrated deposited rate above 50 eV is form-factor independent at the sub-percent level—turning F off shifts it by 0.38%—so reactor CEvNS on Ge sits in the form-factor-insensitive regime and the exact skin parameters are not a systematic driver.

3. Natural-germanium isotope table

The fold retains each stable isotope’s own weak charge, nuclear mass, kinematic endpoint, and number abundance rather than a lumped- A approximation, which would smooth the five-step endpoint structure and bias the near-threshold and near-endpoint bins. Table II lists the five isotopes, with $M_i = A_i \times 931.494 \text{ MeV}$ and $T_{\text{max}}^{(i)}$ at the $E_\nu = 10 \text{ MeV}$ reactor endpoint; the per-isotope target density is $N_{T,i} = x_i \times 8.29 \times 10^{24} \text{ kg}^{-1}$ ($\sum_i N_{T,i} = 8.29 \times 10^{24} \text{ kg}^{-1}$). The five distinct endpoints (2824–3066 eV) produce the stepped superposition seen at the high-recoil edge of dR/dT .

The ^{73}Ge nuclear spin contributes a spin-dependent axial term that is $\sim 1/N^2$ suppressed and negligible for

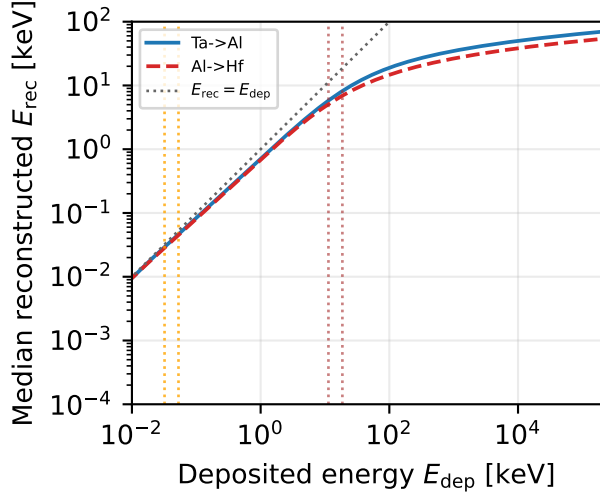


FIG. 3. True-to-reconstructed energy mapping: the median non-paralyzable response $E_{\text{rec}}(E_{\text{dep}})$ for both trapping designs (Ta $\rightarrow$ Al solid, Al $\rightarrow$ Hf dashed), with the on-spot saturation-onset (orange) and whole-array-plateau (brown) deposit scales and the linear-calibration reference $E_{\text{rec}} = E_{\text{dep}}$ (dotted). Below the onset the mapping is linear (the reconstructed energy estimates the deposit); above it the reconstructed energy rolls onto the bandwidth-limited saturation plateau. Nothing below 10 eV is displayed.

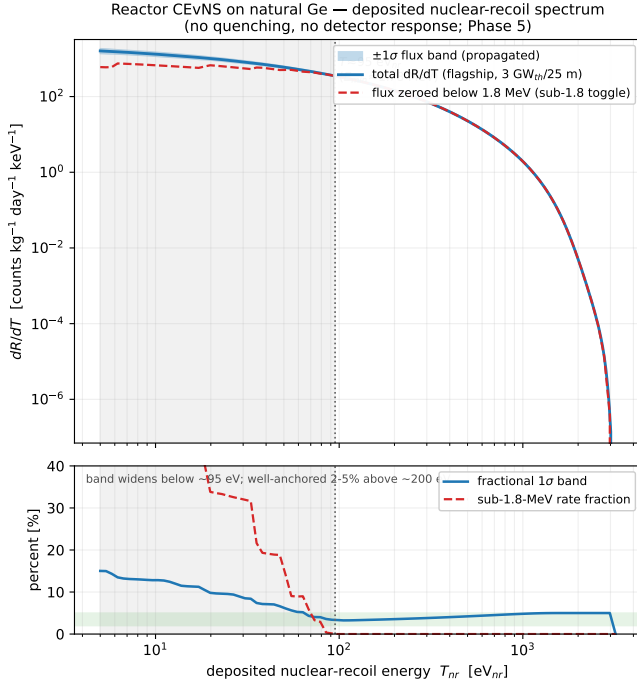


FIG. 4. Deposited nuclear-recoil differential CEvNS rate dR/dT on natural germanium at 3 GW_{th} and 25 m, with the split flux-uncertainty band and the Billard-benchmark reproduction.

TABLE II. The five stable natural-germanium isotopes ($Z = 32$) entering the CEvNS fold of Eq. (6): neutron number N_i , number abundance x_i (IUPAC/CIAAW representative values, $\sum_i x_i = 1$), nuclear mass $M_i = A_i \times 931.494$ MeV, and kinematic endpoint $T_{\text{max}}^{(i)}$ at the $E_\nu = 10$ MeV reactor endpoint.

Isotope	N_i	x_i	M_i (GeV)	$T_{\text{max}}^{(i)}$ (eV)
^{70}Ge	38	0.2057	65.13	3066
^{72}Ge	40	0.2745	66.99	2981
^{73}Ge	41	0.0775	67.93	2940
^{74}Ge	42	0.3650	68.86	2901
^{76}Ge	44	0.0773	70.72	2824

the coherent rate; we state it but do not model it.

Appendix B: Response-matrix Monte Carlo and convergence

This appendix supplies the microphysical detail deferred from Sec. V: the tunneling-burst model that turns a per-sensor share into an event train, the single-constant calibration of the count-integral estimator, the Monte-Carlo construction and convergence of the response matrix $R(E_{\text{rec}} | E_{\text{dep}})$, and the parameter scan that turns the crossover into a band.

EMG tunneling-burst model. On each sensor the trapped-quasiparticle population set by a share E_s drives a two-exponential pulse [Ref. [8], Eq. 4], $N_{\text{qp}}(t) = \tau_{\text{qp}} N_{\text{qp}} g(t)$ with unit-area shape $g(t) = (e^{-t/\tau_{\text{qp}}} - e^{-t/\tau_{\text{inj}}})/(\tau_{\text{qp}} - \tau_{\text{inj}})$, so the instantaneous tunneling rate is $\Gamma_{\text{in}}(t) = (K/V_{\text{tr}})N_{\text{qp}}(t)$. Here τ_{qp} is the trapped-quasiparticle lifetime, τ_{inj} the injection time, K the tunneling-rate constant, and V_{tr} the trap volume. The tunneling events form a burst with exponentially-modified-Gaussian (EMG) arrival times, realized with the `QuasiparticleBurstModel` template of Ref. [8]; its EMG parameters (τ, μ, σ) are fixed once per design so that the pdf peak reproduces $\max_t g(t) = p/\tau_{\text{qp}}$ (recorded peak factors $p = 0.25$ for Al, 0.13 for Hf). The Poisson mean supplied to the template is the expected number of *tunneling events*,

$$n_{\text{ev}} = \int_0^\infty \Gamma_{\text{in}}(t) dt = \frac{K \tau_{\text{qp}}}{V_{\text{tr}}} N_{\text{qp}}, \quad (\text{B1})$$

a dimensionless count that is *not* the trapped population N_{qp} : the two differ by the prefactor $K\tau_{\text{qp}}/V_{\text{tr}} \simeq 0.03$ (Al) and $\simeq 0.008$ (Hf). Feeding N_{qp} instead of n_{ev} would mis-scale the saturation onset by these factors, so Eq. (B1) is enforced everywhere the burst is realized.

Count-integral calibration. Each realized event train is censored by the non-paralyzable dead-time model [Eq. (3)] to a per-sensor observed count, and the array sum N_{obs} is mapped to energy by $E_{\text{rec}} = C N_{\text{obs}}$ [Eq. (16)]. The constant C (eV per event) is a *single*

global value per design, fixed once on a deeply unsaturated deposit ($E_{\text{dep}} = 0.01$ eV, no sensor near the ceiling) by requiring $dE_{\text{rec}}/dE_{\text{dep}} = 1$, i.e. calibrating the reconstructed energy to estimate the deposited energy; this absorbs the physical yield fraction ε into C , exactly as calibrating against a known line would. Because the unsaturated summed count is exactly proportional to E_{dep} , one point fixes C ; it is never retuned per bin. The recovery of $E_{\text{rec}} \approx E_{\text{dep}}$ at low energy is therefore a calibration-consistency check (verified to $< 1\%$ for both designs), while the deep-saturation plateau is the genuinely predictive content: with censoring disabled the estimator returns exactly E_{dep} at all energies, so the plateau is the modeled bandwidth ceiling and not a numerical artifact.

Response-matrix Monte Carlo and convergence. Each E_{dep} column of R is built by histogramming $N_s = 5000$ forward realizations of Eq. (16), on a grid of 584 deposits sampled uniformly in $\log E_{\text{dep}}$ from 10.14 eV to 197 MeV ($\simeq 80$ bins per decade), the shared grid on which Sec. VI folds the deposit spectra. The $\sim \pi r^2$ on-spot sensors are realized individually; the $\sim 10,287$ identical off-spot sensors are drawn as one aggregate multinomial ensemble per realization rather than looped. Columns normalize to unity to 10^{-12} . The per-cell relative Monte-Carlo error is Poisson,

$$\epsilon_{jk} = \frac{1}{\sqrt{N_{jk}}} = \frac{1}{\sqrt{N_s R_{jk}}}, \quad (\text{B2})$$

with N_{jk} the count in cell (j, k) ; the well-populated (peak) cell reaches $\epsilon \lesssim 1.8\%$ at $N_s = 5000$ with clean $1/\sqrt{N_s}$ scaling, and every column is reproducible across processes via deterministic `crc32` sub-seeding. One honest split is required: where the event count is feasible

($n_{\text{ev}} \leq 2000$) the burst is realized with the full EMG template, but in deep saturation the train would carry $\mathcal{O}(10^8)$ events and is infeasible to materialize, so there the registered count is drawn Poisson about the analytic censored mean. In that regime the relative spread of the summed E_{rec} is $\sim 10^{-3}$ —the column is effectively a delta—so the approximation is immaterial to R ; the per-class mean is pinned to the validated analytic estimator in both branches, leaving no kink at the split.

Crossover-band scan. The on-spot sensor saturates first, at the per-sensor onset share E_{onset} ($\simeq 1.27$ eV for Ta→Al, $\simeq 0.77$ eV for Al→Hf); inverting the sharing partition gives the crossover deposit energy $E_{\text{dep}}^{\text{cross}}$ of Eq. (17). Scanning the low-confidence sharing parameters over $f_{\text{prompt}} \in [0.1, 0.5]$ and $r \in [1, 5]$ spreads this from ~ 8 eV to ~ 0.93 keV (Ta→Al) and ~ 4.8 eV to ~ 0.57 keV (Al→Hf) about the default $\sim 52.9/32.1$ eV, the f_{prompt} -dominated band quoted in Sec. V. Two limiting anchors bound it: the equal-split limit $f_{\text{prompt}} \rightarrow 0$, where every sensor shares equally and all saturate together, and the whole-array plateau at the default f_{prompt} , where even the diffuse share reaches onset,

$$\begin{aligned} E_{\text{dep}}^{\text{equal-split}} &= E_{\text{onset}} N_{\text{sens}}, \\ E_{\text{dep}}^{\text{plateau}} &= \frac{E_{\text{onset}} N_{\text{sens}}}{1 - f_{\text{prompt}}}, \end{aligned} \quad (\text{B3})$$

with $N_{\text{sens}} \simeq 10,300$. These evaluate to ~ 13.1 keV (equal-split) and ~ 18.6 keV (plateau) for Ta→Al, and ~ 7.9 keV and ~ 11.3 keV for Al→Hf. The Al→Hf design, with the lower per-sensor onset, saturates before Ta→Al at every point of the scan.

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